04. The Age of Industrialisation

Session 01 –

01. What is meant by proto-industrialisation? How did it affect the rural peasants and artisans?
    - Even before factories began to dot the landscape in England and Europe, there was large-scale industrial production for an international market. This was not based on factories. This phase of industrialisation is referred to as proto-industrialisation.
    - In the seventeenth and eighteenth centuries, merchants from the towns in Europe began moving to the countryside, supplying money to market.
    - It was a time when open fields were disappearing and commons were being enclosed. Cottagers and poor peasants who had earlier depended on common lands for their survival, gathering their firewood, berries, vegetables, hay and straw, had to now look for alternative sources of income.
    - Many had tiny plots of land which could not provide work for all members of the household. So when merchants came around and offered advances to produce goods for them, peasant households eagerly agreed.
    - They could remain in the countryside and continue to cultivate their small plots. Income from proto-industrial production supplemented their shrinking income from cultivation. It also allowed them a fuller use of their family labour resources.
02. Who is Stapler and Fuller?
    Stapler – A person who 'staple' or sorts wool according to its fibre.
    Fuller – A person who 'fulls' – that is, gathers – cloth by pleating.
03. What is carding?
    The process in which fibres, such as cotton or wool, are prepared prior to spinning.
04. Which city is known as the finishing centre?
    London
05. What was the symbol of the new era in England in the late eighteenth century?
    The first symbol of the new era was cotton.
06. Which industries were the dynamic industries of England during the earliest phase of industrialisation?
    The most dynamic industries in Britain were clearly cotton and metals.
07. The modern industrialisation could not marginalise the traditional industries in England. Justify the statement.

    OR

    What was the phase of industrialisation in England?
    - Even at the end of the nineteenth century, less than 20 per cent of the total workforce was employed in technologically advanced industrial sectors. Textiles was a dynamic sector, but a large portion of the output was produced not within factories, but outside, within domestic units.
    - Ordinary and small innovations were the basis of growth in many non-mechanised sectors such as food processing, building, pottery, glass work, tanning, furniture making, and production of implements.

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- technological changes occurred slowly. They did not spread dramatically across the industrial landscape. New technology was expensive and merchants and industrialists were cautious about using it. The machines often broke down and repair was costly.

08. Who improved the steam engine produced by Newcomen?
James Watt improved the steam engine produced by Newcomen and patented the new engine in 1781.

Session 02 –

01. Why did some industrialists in nineteenth century Europe prefer hand labour to machine?
- Machines were costly, ineffective and difficult to repair and required huge capital investment.
- Labour was available at low wages at that period of time.
- In seasonal industries, only seasonal labour was required.
- Market demands of a variety of designs and colour at specific type could not be fulfilled by machine-made clothes. It could be done by human skill only.
- In Victorian age, the aristocrats and other upper-class people preferred articles made by hand only.

02. What was spinning jenny?
Devised by James Hargreaves in 1764, this machine speeded up the spinning process and reduced labour demand. By turning one single wheel a worker could set in motion a number of spindles and spin several threads at the same time.

Session 03 –

01. Before the age of industrialisation, silk and cotton goods from India dominated the international market in textiles.

OR

What led to the rise of Bombay and Calcutta ports in the nineteenth century? Explain.
Before the age of machine industries, silk and cotton goods from India dominated the international market in textiles. Coarser cottons were produced in many countries, but the finer varieties often came from India. Armenian and Persian merchants took the goods from Punjab to Afghanistan, eastern Persia and Central Asia. Bales of fine textiles were carried on camel back via the north-west frontier, through mountain passes and across deserts. A vibrant sea trade operated through the main pre-colonial ports. Surat on the Gujarat coast connected India to the Gulf and Red Sea Ports; Masulipatam on the Coromandel coast and Hoogly in Bengal had trade links with Southeast Asian ports.

02. What role did the Indian merchants and bankers played in the export trade in the early eighteenth century?
A variety of Indian merchants and bankers were involved in this network of export trade – financing production, carrying goods and supplying exporters. Supply merchants linked the port towns to the inland regions. They gave advances to weavers, procured the woven cloth from weaving villages, and carried the supply to the ports. At the port, the big shippers and

export merchants had brokers who negotiated the price and bought goods from the supply merchants operating inland.

03. Why the trade network controlled by Indian merchants breakdown by 1750s?

The European companies gradually gained power – first securing a variety of concessions from local courts, then the monopoly rights to trade. This resulted in a decline of the old ports of Surat and Hoogly through which local merchants had operated. Exports from these ports fell dramatically, the credit that had financed the earlier trade began drying up, and the local bankers slowly went bankrupt. In the last years of the seventeenth century, the gross value of trade that passed through Surat had been Rs 16 million. By the 1740s it had slumped to Rs 3 million.

04. Explain way through which British manufacturers attempted to take over Indian market.

- It proceeded to develop a system of management and control that would eliminate competition, control costs, and ensure regular supplies of cotton and silk goods.
- It appointed a paid servant called the gomastha to supervise weavers, collect supplies, and examine the quality of cloth.
- It introduced the system of advances. Once an order was placed, the weavers were given loans to purchase the raw material for their production. Those who took loans had to hand over the cloth they produced to the gomastha. They could not take it to any other trader.

05. Explain the miserable conditions of Indian weavers during the EIC regime in the eighteenth century.

- Earlier supply merchants had very often lived within the weaving villages, and had a close relationship with the weavers, looking after their needs and helping them in times of crisis. The new gomasthas were outsiders, with no long-term social link with the village.
- They acted arrogantly, marched into villages with sepoys and peons, and punished weavers for delays in supply – often beating and flogging them.
- The weavers lost the space to bargain for prices and sell to different buyers: the price they received from the Company was miserably low and the loans they had accepted tied them to the Company.

06. Describe any three major problems faced by Indian cotton weavers in the nineteenth century.

OR

Describe three reasons for the decline of the textile exports from India in the nineteenth century.

- As cotton industries developed in England, industrial groups began worrying about imports from other countries. They pressurised the government to impose import duties on cotton textiles so that Manchester goods could sell in Britain without facing any competition from outside.
- At the same time industrialists persuaded the East India Company to sell British manufactures in Indian markets as well. Exports of British cotton goods increased dramatically in the early nineteenth century. By 1850 cotton piece-goods constituted over 31 per cent of the value of Indian imports; and by the 1870s this figure was over 50 per cent.
- Cotton weavers in India thus faced two problems at the same time: their export market collapsed, and the local market shrank, being glutted with Manchester imports. Produced

by machines at lower costs, the imported cotton goods were so cheap that weavers could not easily compete with them.

- By the 1860s, weavers faced a new problem. They could not get sufficient supply of raw cotton of good quality. When the American Civil War broke out and cotton supplies from the US were cut off, Britain turned to India. As raw cotton exports from India increased, the price of raw cotton shot up. Weavers in India were starved of supplies and forced to buy raw cotton at exorbitant prices. In this, situation weaving could not pay.
- Factories in India began production, flooding the market with machine-goods.

Session 04 –

01. How did the industries develop in India in the second-half of the nineteenth century? Explain.

OR

Describe the contribution of the early industrialists of India in shaping the industrial development of India.

- The British in India began exporting opium to China and took tea from China to England. Many Indians became junior players in this trade, providing finance, procuring supplies, and shipping consignments. Having earned through trade, some of these businessmen had visions of developing industrial enterprises in India.
- In Bengal, Dwarkanath Tagore made his fortune in the China trade before he turned to industrial investment, setting up six joint-stock companies in the 1830s and 1840s.
- In Bombay, Parsis like Dinshaw Petit and Jamsetjee Nusserwanjee Tata who built huge industrial empires in India, accumulated their initial wealth partly from exports to China, and partly from raw cotton shipments to England.
- Seth Hukumchand, a Marwari businessman who set up the first Indian jute mill in Calcutta in 1917, also traded with China. So did the father as well as grandfather of the famous industrialist G.D. Birla.
- Some merchants from Madras traded with Burma while others had links with the Middle East and East Africa. There were yet other commercial groups, but they were not directly involved in external trade. They operated within India, carrying goods from one place to another, banking money, transferring funds between cities, and financing traders. When opportunities of investment in industries opened up, many of them set up factories.

02. Who were the jobbers and explain their functions?

Industrialists usually employed a jobber to get new recruits. Very often the jobber was an old and trusted worker. He got people from his village, ensured them jobs, helped them settle in the city and provided them money in times of crisis. The jobber therefore became a person with some authority and power. He began demanding money and gifts for his favour and controlling the lives of workers.

Session 05 –

01. By the first decade of the twentieth century a series of changes affected the pattern of industrialisation of India. Explain.

- As the swadeshi movement gathered momentum, nationalists mobilised people to boycott foreign cloth. Industrial groups organised themselves to protect their collective interests, pressurising the government to increase tariff protection and grant other concessions.
- From 1906, moreover, the export of Indian yarn to China declined since produce from Chinese and Japanese mills flooded the Chinese market. So industrialists in India began shifting from yarn to cloth production.
- With British mills busy with war production to meet the needs of the army, Manchester imports into India declined. Suddenly, Indian mills had a vast home market to supply.
- As the war prolonged, Indian factories were called upon to supply war needs: jute bags, cloth for army uniforms, tents and leather boots, horse and mule saddles and a host of other items.
- New factories were set up and old ones ran multiple shifts.

02. Why did industrial production in India increased during the First World War? Explain.

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- As the war prolonged, Indian factories were called upon to supply war needs: jute bags, cloth for army uniforms, tents and leather boots, horse and mule saddles and a host of other items.
- New factories were set up and old ones ran multiple shifts.
- Many new workers were employed and everyone was made to work longer hours. Over the war years industrial production boomed.

03. Small-scale production continued to pre-dominate in India. Why?

OR

Handicraft production expanded in the twentieth century. Explain.

- Small-scale production continued to predominate – Certain groups of weavers were in a better position than others to survive the competition with mill industries. Amongst weavers some produced coarse cloth while others wove finer varieties. The coarser cloth was bought by the poor and its demand fluctuated violently.
- In some instances, handicrafts production actually expanded in the twentieth century – This was partly because of technological changes. Handicrafts people adopt new technology if that helps them improve production without excessively pushing up costs. So, by the second decade of the twentieth century we find weavers using looms with a fly shuttle.

04. What was the fly shuttle?

It is a mechanical device used for weaving, moved by means of ropes and pullies. It places the horizontal threads (called the weft) into the vertical threads (called the warp). The invention of the fly shuttle made it possible for weavers to operate large looms and weave wide pieces of cloth.

Session 06 –

01. Explain any five ways by which new market and consumers were created.

- Advertisements made products appear desirable and necessary. They try to shape the minds of people and create new needs.

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- The label was needed to make the place of manufacture and the name of the company familiar to the buyer. When buyers saw 'MADE IN MANCHESTER' written in bold on the label, they were expected to feel confident about buying the cloth.
- Images of Indian gods and goddesses regularly appeared on these labels. The imprinted image of Krishna or Saraswati was also intended to make the manufacture from a foreign land appear somewhat familiar to Indian people.
- By the late nineteenth century, manufacturers were printing calendars to popularise their products. Unlike newspapers and magazines, calendars were used even by people who could not read. They were hung in tea shops and in poor people's homes just as much as in offices and middle-class apartments.

Source: Binita Jain ma'am – History teacher, BMSSS
NCERT textbook

Garima Trivedi

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